

ARS-Avatar: Animatable and Relightable Surfel Avatars with Learnable Ambient Occlusion

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<https://arsavatar.github.io/>

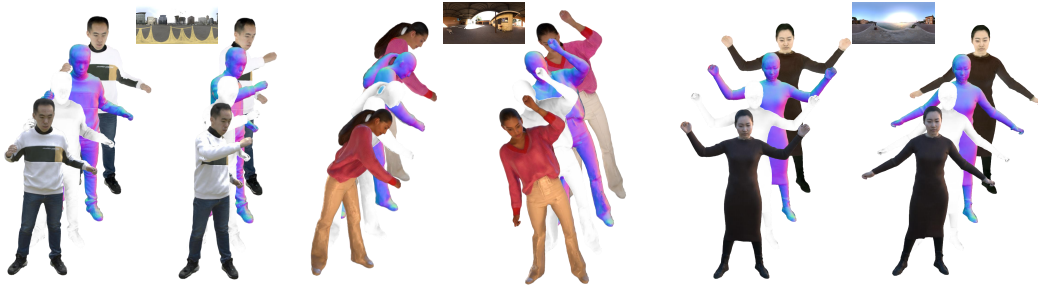


Figure 1. **ARS-Avatar** creates animatable and relightable human avatars with surfels. Given novel poses and illuminations, it reconstructs high-fidelity *appearance*, *geometry*, *BRDF materials*, and *optimizable occlusion* for realistic animation and relighting.

Abstract

Creating animatable and relightable human avatars from multi-view images remains challenging, as pose-dependent deformation, materials, and light visibility are intrinsically coupled in images. In this paper, we present **ARS-Avatar**, a novel method using surfel representation for high-quality, animatable, and relightable human avatars from multi-view images captured under unknown illumination. We first extract deformation priors from the template mesh and leverage as additional details beyond driving poses to facilitate faithful estimation of surfel attributes and reconstruction of animatable avatar. To support relighting, the deferred shading is employed to estimate BRDF materials. We further introduce a differentiable screen-space ambient occlusion formulation that enables gradient-based optimization of body-part specific occlusion radii through finite differences, providing an efficient approximation of light visibility that can be jointly optimized with the avatar. Extensive experiments demonstrate that ARS-Avatar achieves high-fidelity appearance reconstruction and physically-based material estimation, while enabling realistic animation and relighting un-

der novel poses and illuminations.

1. Introduction

Creating animatable and relightable human avatars remains a fundamental problem in computer graphics, with applications in embodied AI (EAI), VR/AR, virtual try-on, and telepresence. Traditional methods rely on high-quality 3D scans and calibrated illumination [14, 16], however modeling such avatars solely from multi-view images captured under unknown illumination remains highly ill-posed, as geometry, material properties, and scene illumination are tightly coupled.

Recent advances in neural rendering, particularly NeRF [41] and 3DGS [29], have substantially improved high-fidelity human avatar reconstruction from multi-view images. Although NeRF-based methods [37, 57, 59, 61] achieve impressive quality, expensive volume rendering limits practicality, motivating efficient explicit representations [26]. Building on 3DGS, recent methods [12, 26, 33, 54, 63, 66] reconstruct animatable and relightable avatars by combining primitives with templates [40, 45] and physically-based rendering [5, 47]. However, existing methods rarely fully consider deformation priors in opti-

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mization. This limitation is particularly evident for clothed humans, where a position map or driving pose alone usually not fully represent rich pose-dependent deformation. Furthermore, although visibility estimation has been studied in meshes [10, 63] and volume rendering [53, 59], extending it to discrete Gaussian surfels is non-trivial. These limitations jointly constrain the fidelity of animation and relighting.

Therefore, we present **ARS-Avatar**, an animatable and relightable human avatar represented by GaussianSurfels [13] that achieves photorealistic rendering under arbitrary poses and novel illuminations. Specifically, we employ SMPL-X [45] to construct the canonical avatar and reconstruct appearance, geometry, and BRDF materials under a driving pose. The driving pose is encoded into a position map [35] and fed into StyleUNets to estimate surfel attributes. We further extract two deformation priors from the template mesh, i.e., the deformation displacement and the deviation from the parametric body model, both of which reflect local deformation complexity. The extracted priors are incorporated into the estimation process, yielding more faithful pose-dependent details. Furthermore, we employ screen-space ambient occlusion (SSAO) as an efficient approximation and incorporate it into the optimization via a finite-difference formulation with respect to the occlusion radius. Each human body part has an initial occlusion radius and a learnable radius offset, and the radii are optimized via finite differences. This formulation enables end-to-end optimization of occlusion while maintaining training efficiency.

In summary, our contributions include:

- A extraction method of deformation priors that then incorporate into estimation of surfel attributes, helping faithful pose-dependent reconstruction.
- A differentiable occlusion formulation based on finite differences enables physically plausible estimation of occlusion and efficient optimization of occlusion radius.
- Extensive experiments on two datasets demonstrate that **ARS-Avatar** achieves state-of-the-art reconstruction quality and enables photorealistic animation and relighting.

2. Related Work

2.1. Animatable Human Avatar Modeling

Early methods rely on explicit meshes [2, 3, 6, 60] and template body models such as SMPL [40] and SMPL-X [45] to reconstruct and deform avatars. Later methods introduce learnable deformation fields to capture pose-dependent deformations [17, 58]. More recent NeRF-based methods [9, 31, 34, 39, 52, 68, 69] model complex geometry and appearance for continuous, high-fidelity reconstruction and novel pose animation, but ray sampling limits efficiency. Explicit 3DGS [29] has thus emerged as an effective representation for animatable avatars [19, 20, 25, 30,

32, 35, 43, 46, 67, 71]. Despite efficient rendering, modeling complex clothed humans remains challenging, as conditioning solely on the driving pose does not fully capture rich pose-dependent deformation. To address this, we propose a deformation prior extraction and incorporation method for more faithful pose-dependent reconstruction.

2.2. Relightable Human Avatar Modeling

Relightable avatar modeling is an inverse rendering problem that disentangles geometry, materials, and illumination. Early works use neural networks to relight single images [24, 27, 42, 49, 56, 70], but provide limited support for animation. Chen and Liu [11] pioneered integrating physically-based rendering [50] into NeRF for human geometry and materials. Subsequent methods [22, 59] improve appearance modeling for detailed relighting, while others [7, 37, 53, 61] use explicit light transport for accurate intrinsic decomposition and physically plausible relighting. Recent Gaussian avatars combine explicit primitives with physically based material and illumination models [12, 26, 33, 48, 54, 62, 63], demonstrating efficient, high-fidelity relightable avatars. In contrast, our work explores Gaussian Surfels [13] as the base representation, leveraging their surface-aligned representation for high-fidelity appearance and geometry.

2.3. Visibility and Occlusion Estimation

Visibility is crucial in physically-based rendering, determining light transport between light and surface points. A common approach computes visibility via ray tracing on meshes [10], but requires high-quality triangle meshes. NeRF-based methods incorporate visibility into MLPs [59, 61]; Lin *et al.* [37] divide the body into parts for more accurate self-occlusion. Recently, visibility for explicit Gaussians has gained attention [15]. One strategy combines Gaussians with a mesh and uses mesh visibility [63]. Liang *et al.* [36] model visibility using ambient occlusion (AO) and encode it into spherical harmonics for efficient querying, later extended to dynamic avatars by Jiang *et al.* [26]. Chen *et al.* [8] adopts screen-space ambient occlusion (SSAO) to improve occlusion efficiency. However, these approaches treat visibility as post-computed or detached, making joint optimization with the avatar difficult. In contrast, our method formulates SSAO with respect to the occlusion radius through finite differences, enabling end-to-end occlusion optimization while maintaining the efficiency required for practical avatar reconstruction.

3. Method

3.1. Overview

Given multi-view images of a clothed human, we aim to create an animatable and relightable avatar. As shown in

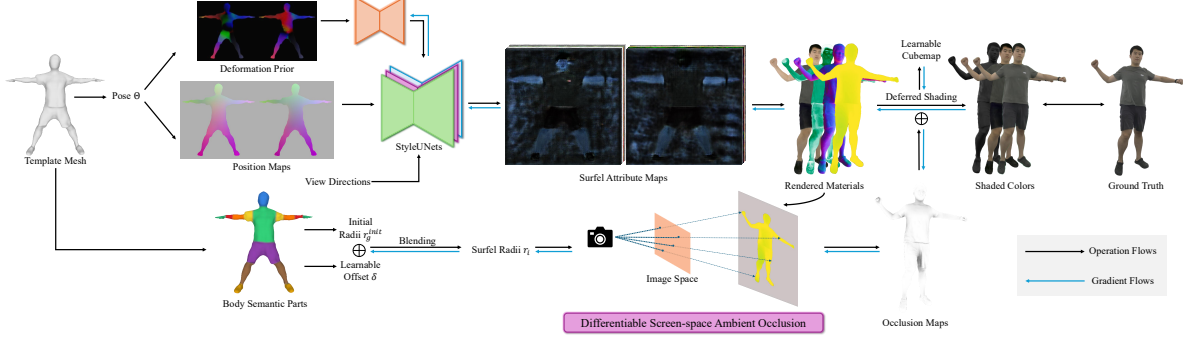


Figure 2. **An illustration of our method.** Given a canonical template mesh, we first deform it into the observation space under a driving pose Θ to generate position maps and extract deformation prior maps based on deformation priors. Both are fed into multiple StyleUNets to estimate attributes of each surfel. We then employ deferred shading to estimate material properties for relighting under novel illumination. Finally, we introduce a differentiable screen-space ambient occlusion (SSAO) module that efficiently estimates occlusion and enables joint optimization with the avatar representation during training.

Fig. 2, we represent the avatar with Gaussiansurfels [13]. Following Li [35], we estimate pose-dependent surfel attributes from position maps using StyleUNets [51]. To better capture the rich deformation features, we extract deformation priors from the template and incorporate them into estimation (Sec. 3.2). We further employ deferred shading to estimate BRDF materials (Sec. 3.3). Light visibility is approximated as screen-space ambient occlusion (SSAO), made differentiable via a finite-difference formulation with respect to the occlusion radius, enabling efficient radius optimization while incorporating visibility into training (Sec. 3.4). Finally, **ARS-Avatar** creates animatable and relightable human avatars under any target pose, viewpoint, and illumination.

3.2. Surfel Estimation with Deformation Priors

We initialize the canonical avatar with surfels $\mathcal{G}$, each parameterized by position $\mu_c \in \mathbb{R}^3$, rotation $\mathbf{r}_c \in \mathbb{R}^4$, scale $s \in \mathbb{R}^3$, opacity $\mathbf{o} \in \mathbb{R}^1$, and directional color $\mathbf{c} \in \mathbb{R}^3$. For physically-based shading, we additionally define albedo $\mathbf{a} \in \mathbb{R}^3$, and roughness $\rho \in \mathbb{R}^1$. Given a coarse template mesh and driving pose Θ , we follow Li *et al.* [35] and rasterize template vertices x_c colored by posed coordinates x_o from front and back views to obtain position maps:

$$\mathbf{M}_{pos}^f(\Theta), \mathbf{M}_{pos}^b(\Theta) = \text{Raster}(x_c, x_o; \Theta), \quad (1)$$

where $\text{Raster}(\cdot; \Theta)$ rasterizes the template mesh under pose Θ from front and back views. For deformation priors, we use two vertex-wise priors: a) the distance before and after deformation $\mathbf{d}$, and b) the distance to the nearest SMPL-X [45] vertex $\mathbf{s}$, both positively correlated with deformation complexity. We render them similarly to obtain deformation prior maps:

$$\mathbf{M}_{def}^f(\Theta), \mathbf{M}_{def}^b(\Theta) = \text{Raster}(\mathbf{x}_c, \{\mathbf{d}, \mathbf{s}\}; \Theta), \quad (2)$$

where $\{\cdot\}$ denotes concatenate operation and we discuss it in the appendix. We encode these maps with a neural network to generate deformation feature maps,

$$f = \mathcal{F}(\mathbf{M}_{def}^f(\Theta), \mathbf{M}_{def}^b(\Theta)), \quad (3)$$

where $\mathcal{F}(\cdot)$ is the neural network. The feature maps are further used as auxiliary to incorporate with position maps into StyleUNets to estimate surfel attributes,

$$\mu_c, \mathbf{r}_c, \mathbf{s}, \mathbf{o}, \mathbf{c}, \mathbf{a}, \rho = \mathcal{S}(\mathbf{M}_{pos}^f, \mathbf{M}_{pos}^b; f, \mathcal{V}), \quad (4)$$

where $\mathcal{V}$ denotes view feature map employed in [35] and $\mathcal{S}(\cdot)$ is the StyleUNets. We further deform canonical surfels to observation space via Linear Blend Skinning (LBS),

$$\mu_o = \mathcal{R}(\mu_c) + \mathcal{T}, \quad \mathbf{r}_o = \mathcal{R}(\mathbf{r}_c), \quad (5)$$

where $\mathcal{R}$ and $\mathcal{T}$ are rotation and translation from skinning weights. The covariance matrix is:

$$\Sigma_o = \mathbf{r}_o \mathbf{s} \mathbf{s}^T \mathbf{r}_o^T, \quad (6)$$

finally, we render the avatar to obtain the rendered color $\hat{C}$, depth $\hat{D}$, normal $\hat{N}$, albedo $\hat{A}$, roughness $\hat{H}$, and position $\hat{P}$,

$$\hat{C}, \hat{D}, \hat{N}, \hat{A}, \hat{H}, \hat{P} = \text{Splat}(\mathbf{c}, \mathbf{a}, \rho; \mu_o, \Sigma_o, \mathbf{o}), \quad (7)$$

where $\text{Splat}(\cdot; \cdot)$ is EWA splatting [72] derived from [13] and extended to render additional feature maps. We further extend the formulation in Sec. 3.4.

3.3. Material Estimation with Deferred Shading

We follow the rendering equation [47] for physically-based shading,

$$L_o(\omega_o) = \int_{\Omega} L_i(\omega_i) f_r(\omega_i, \omega_o) (\omega_i \cdot \mathbf{n}) d\omega_i, \quad (8)$$

where $L_o(\omega_o)$ and $L_i(\omega_i)$ are outgoing radiance in direction $\omega_o = P_{\text{cam}} - \hat{P}$ and incident light from ω_i . The incident light is represented by a learnable HDR cube map with the shape $6 \times 512 \times 512$. We use the Disney shading model [5] with albedo $\mathbf{a}$, and roughness ρ as BRDF materials. We parameterize the BRDF at each pixel. We decompose the BRDF term f_r into diffuse f_{diff} and specular f_{spec} , the shaded color is,

$$\hat{C}_{\text{shade}} = C_{\text{diff}} + C_{\text{spec}}, \quad (9)$$

we discuss the definition of above 2 terms in the appendix. Combining with ambient occlusion discussed in Sec. 3.4, the physically-based shading color is:

$$\hat{C} = \mathcal{O} \odot \hat{C}_{\text{shade}}, \quad (10)$$

where $\odot$ denotes element-wise multiplication. The learnable cube map L_i is optimized jointly with surfels to estimate incident illumination from observations.

3.4. Differentiable Screen-space Ambient Occlusion

We approximate light visibility with screen-space ambient occlusion (SSAO) [4], which estimates local visibility from rendered depth $\hat{D}$ and normal $\hat{N}$. For each pixel p ,

$$\text{SSAO}(p) = \frac{1}{N} \cdot \sum_{i=1}^N V(p, \omega_i), \quad (11)$$

where $V(p, \omega_i) \in [0, 1]$ is visibility along sampled direction ω_i . However, standard SSAO is not directly amenable to end-to-end optimization, and optimizing its input depth and normal maps would implicitly alter geometry and surface orientation. We therefore introduce a dedicated optimizable path that only adjusts occlusion radius while keeping geometry inputs fixed. We divide the human body into 9 semantic parts, each with an initial radius and a learnable scalar offset. For part g :

$$r_g = r_g^{\text{init}} (1 + \delta_g), \quad g = 1, \dots, 9, \quad (12)$$

the radius associated with each surfel is obtained by blending the radii of its neighboring joints according to the LBS weights:

$$r_i = \sum_{j=1}^J w_{ij} r_{g(j)}, \quad (13)$$

where w_{ij} denotes the LBS weight of surfel i for joint j , and $g(j)$ maps joint j to its corresponding body part.

We generate perturbed radii r_g^+ and r_g^- by adding a perturbation offset $\varepsilon \Delta_g$ to the corresponding body parts, where $\Delta_g \in \{-1, +1\}$ and ε is a small perturbation magnitude. We splat above 3 radii to obtain radius map $\hat{R}, \hat{R}_+, \hat{R}_-$ via

Eq. 7. The final and perturbed occlusion map is computed as:

$$\hat{\mathcal{O}}, \hat{\mathcal{O}}_+, \hat{\mathcal{O}}_- = \mathcal{F}(\hat{N}, \hat{D}; \hat{R}, \hat{R}_+, \hat{R}_-), \quad (14)$$

Therefore, we estimate the gradient via the finite difference formulation,

$$\frac{\partial \hat{\mathcal{O}}}{\partial \hat{R}} \approx \frac{\hat{\mathcal{O}}_+ - \hat{\mathcal{O}}_-}{\hat{R}_+ - \hat{R}_-}, \quad (15)$$

since PyTorch [44] and the EWA splatting [72] are differentiable, the full gradient for radius compensation is:

$$\frac{\partial \mathcal{L}}{\partial \delta} = \frac{\partial \mathcal{L}}{\partial \hat{\mathcal{O}}} \cdot \frac{\partial \hat{\mathcal{O}}}{\partial \hat{R}} \cdot \frac{\partial \hat{R}}{\partial r_i} \cdot \frac{\partial r_i}{\partial \delta_g}, \quad (16)$$

finally, the final shading result is obtained by:

$$\hat{C}_{\text{pbr}} = C_{\text{shade}} \cdot \hat{\mathcal{O}}, \quad (17)$$

we provide a more detailed explanation in the appendix.

3.5. Loss Function and Optimization

We optimize ARS-Avatar using multi-view RGB images C . The training losses combines L1, perceptual, geometry loss, normal deformation, light, BRDF, and regularization:

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_{\text{lips}} + \mathcal{L}_{\text{geo}} + \mathcal{L}_{\text{nd}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{TV}} + \mathcal{L}_{\text{light}}, \quad (18)$$

the geometry loss enforces consistency between rendered normal and depth map. We further propose $\mathcal{L}_{\text{nd}}$ to regularize normal deformation,

$$\mathcal{L}_{\text{nd}} = \frac{1}{|\mathcal{K}|} \sum_{(i,j) \in \mathcal{K}} w_{ij} (1 - |\mathbf{n}_{oi} \cdot \mathbf{n}_{oj}|), \quad (19)$$

where $\mathbf{n}_{oi}$ denotes the observation normal of i -th surfel, and $\mathcal{K}$ contains selected neighboring pairs satisfying the canonical distance constraint. w_{ij} is a spatial weight based on observation space distance,

$$w_{ij} = \exp\left(-\frac{\|\boldsymbol{\mu}_{oi} - \boldsymbol{\mu}_{oj}\|_2^2}{\sigma_i^2}\right), \quad (20)$$

where σ_i is the local distance scale from the farthest neighbor in the K -nearest neighborhood of the i -th surfel.

4. Experiments

Dataset. The experiments are conducted on Avatar-ReX [69] and ActorsHQ [23] datasets. We follow the setting in [35] and select 4 characters from AvatarReX and 2 actors from ActorsHQ dataset for training.

Metric. We employ Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [55], Learned Perceptual Image Patch Similarity (LPIPS) [65], and Frechet Inception Distance (FID) [18] as metrics.



Figure 3. **Illustration of appearance and geometry reconstruction, BRDF materials estimation, and relighting synthesis under training and novel poses.** The second column shows the results of directly fitting Surfel colors, while the third column shows the results obtained using physically-based rendering.

Baseline. We compare radiance reconstruction against 3DGS-Avatar [46], WebAvatar [64], and AnimatableGaussian [35], and further compare physically-based shading and material estimation against MeshAvatar [10] and Relighting4D [11].

Results. We report animation and relighting results of clothed humans created by ARS-Avatar in Fig. 1 and Fig. 3.

4.1. Comparison

Animatable Avatar Reconstruction. We first evaluate the animation capability of RAS-Avatar against state-of-the-art avatar reconstruction methods, including PoseVocab [34],

Table 1. **Quantitative Comparison against state-of-the-art radiance reconstruction baseline on AvatarReX [69] and Actor-sHQ [23] dataset.** *The results of AG [35] is from their paper.

	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
REX	3DA [46]	29.2612	0.9729	0.0389	29.6184
	AG* [35]	30.6143	0.9803	0.0290	<u>13.2417</u>
	WEB [64]	31.6588	<u>0.9809</u>	<u>0.0297</u>	23.1407
	Ours	<u>30.7514</u>	0.9862	0.0318	13.1050
HQ	3DA [46]	30.1744	0.9567	<u>0.0317</u>	16.7763
	AG* [35]	30.3607	0.9682	<u>0.0339</u>	33.4665
	WEB [64]	<u>31.3856</u>	<u>0.9686</u>	0.0324	31.5130
	Ours	31.4785	0.9694	0.0299	<u>18.9706</u>

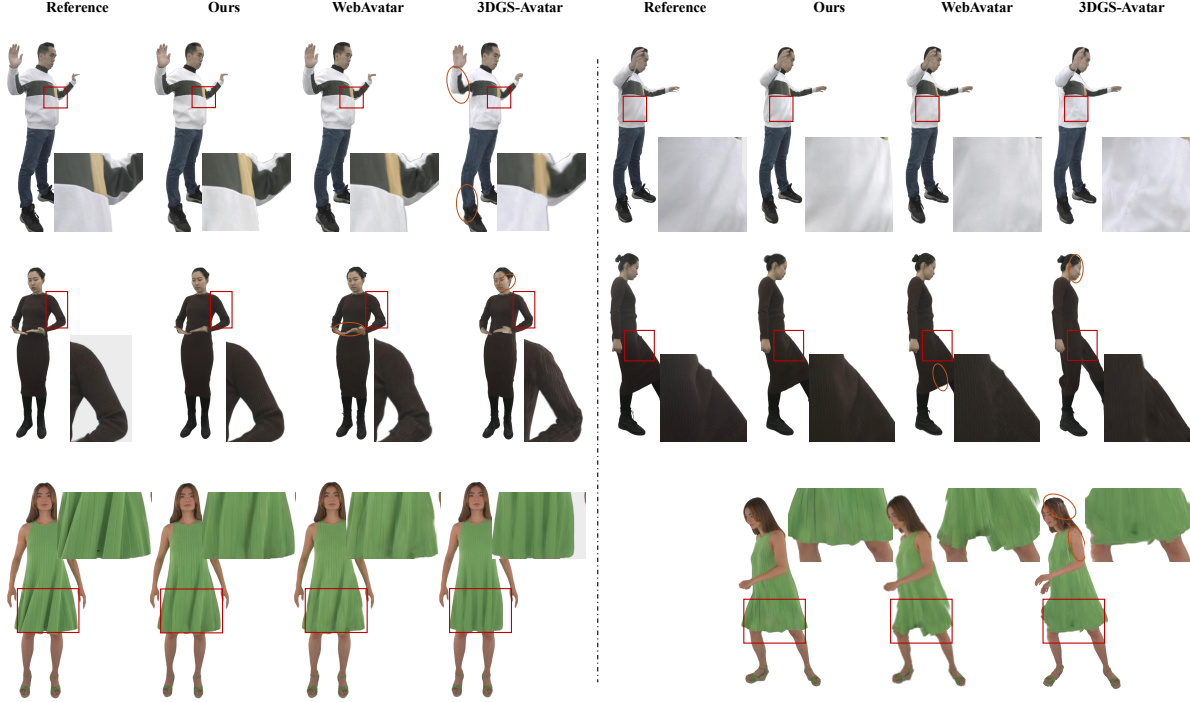


Figure 4. **Qualitative Radiance Comparison** with state-of-the-art human avatar reconstruction methods including 3DGS-Avatar [46] and WebAvatar [64] under testing and novel poses.

Table 2. **Quantitative Comparison** against state-of-the-art PBR reconstruction baseline on AvatarReX [69] and ActorsHQ [23] dataset.

	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
REX	R4D [11]	23.6745	0.9252	0.0974	96.4823
	MA [10]	28.5608	0.9707	0.0377	18.2602
	Ours	30.2703	0.9752	0.0375	12.7732
HQ	R4D [11]	24.1780	0.9457	0.0627	139.2214
	MA [10]	28.7860	0.9510	0.0305	19.7140
	Ours	31.6116	0.9596	0.0261	17.1087

Table 3. **Quantitative Ablation on Deformation Priors (DP).**

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
Ours w/o MPG	30.9457	0.9515	0.0446	32.5386
Ours	31.7459	0.9560	0.0382	33.4016

AnimatableGaussians [33], and WebAvatar [64]. Qualitative and quantitative results are reported in Fig. 4 and Tab. 1. ARS-Avatar achieves superior metric results and high-fidelity visual results under novel poses, especially in regions with complex nonrigid deformation, such as loose garments and articulated body parts. In contrast, existing methods generally rely on rigid deformation and driving

pose information, which may fail to capture the rich information of clothed humans.

Relightable Avatar Reconstruction. We further compare the relighting capability of RAS-Avatar with recent physically based avatar reconstruction methods, including MeshAvatar [10], and Relighting4D [11]. Qualitative and quantitative comparisons are reported in Fig. 5 and Tab. 2, respectively. ARS-Avatar produces more realistic illumination and occlusion effects under novel lighting. Although explicit mesh or neural field methods achieve physically based appearance modeling, their visibility estimation relies on explicit geometry processing or expensive light transport. In contrast, Our introduces a learnable screen-space ambient occlusion formulation, enabling efficient visibility approximation and joint optimization with the avatar. This yields more consistent self occlusion and contact visibility without costly ray tracing or offline computation.

4.2. Ablation Study

We conduct extensive ablation studies to evaluate the effectiveness of our proposed components.

Effect of Deformation Priors. We compare the model without deformation prior with our full model, where pose-dependent surfel attributes are directly predicted from the position map. As shown in Fig. 7, introducing deformation priors improves the reconstruction quality by providing additional spatial guidance for learning pose-dependent off-



Figure 5. Qualitative Physical Comparison with state-of-the-art human avatar reconstruction and relighting methods including meshavatar [10] and relighting4D [11] under testing and novel poses.

sets. This enables the model to better preserve fine-grained pose-dependent details, especially in areas involving large-

range movements, such as the limbs.

Effect of Differentiable Screen-space Ambient Occlu-

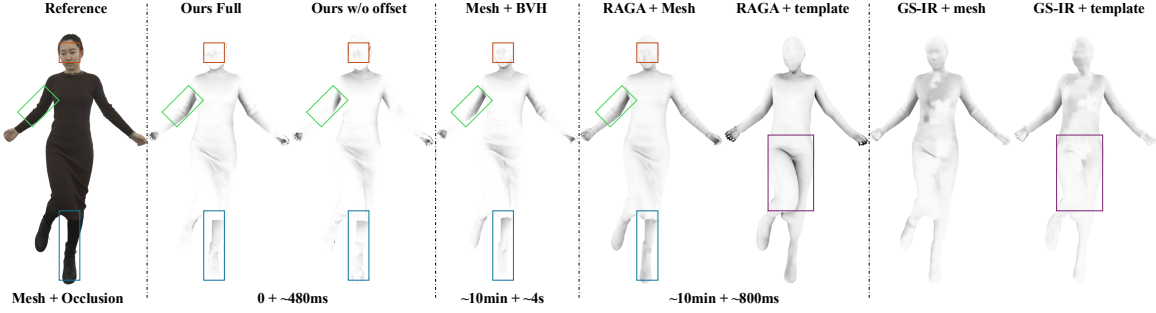


Figure 6. **Qualitative Ablation on Differentiable Screen-space Ambient Occlusion.** We compare occlusion results computed by five variants: (1) our full method; (2) our method without the optimizable radii offsets; (3) computation on mesh extracted by [13] and accelerated by BVH [28]; (4) method proposed by RAGA [63]; (5) method proposed by GS-IR [36]. Our method produces high-quality occlusion results in real time without additional operation. time for mesh extraction and occlusion computation is reported below. Since the results of (5) is abnormal, we omit a detailed time analysis.

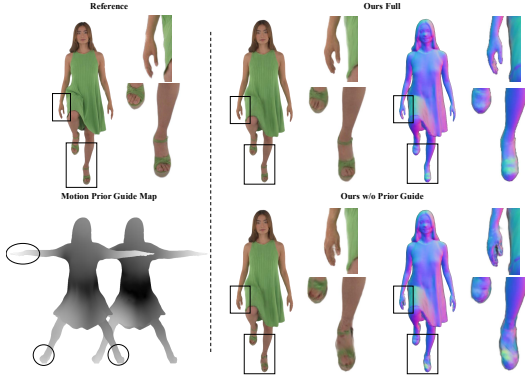


Figure 7. **Qualitative Ablation on Deformation Priors.** Our deformation prior focuses on regions with significant deformation, such as the hands and feet. Incorporating it into surfel estimation yields more refined reconstructions in these regions.

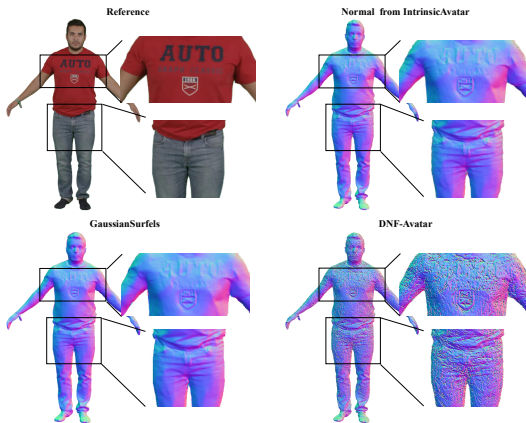


Figure 8. **Qualitative Ablation on Surfel Representation.** Compared with [21], gaussiansurfel [13] representation reconstructs more accurate depth. We convert the depth into normal for clearer comparison. Training process is based on [26] and normal supervision is derived from [53].

ion. We validate the results with 4 variants, which is discussed in Fig. 6. Although fixed SSAO improves the visual fidelity, its performance is limited because the estimated visibility cannot be jointly optimized with the avatar representation. In contrast, our formulation achieves superior results by enabling end-to-end optimization of ambient occlusion while preserving the efficiency of screen-space computation.

Effect of Gaussian Surfel Representation. Unlike conventional 3D Gaussian primitives, Gaussian Surfels explicitly encode surface orientation via surface-aligned anisotropic primitives, providing more reliable geometry for physically based rendering. As shown in Fig. 8, they produce more coherent depth and normal maps with sharper geometric boundaries than conventional Gaussians, which benefits material estimation and visibility modeling and leads to more realistic relighting.

5. Conclusion

We present **ARS-Avatar**, which creates animatable and relightable human avatars from multi-view images. Surfel primitives achieve high fidelity and consistency in appearance and geometry with faster rendering than implicit methods. We further propose a differentiable ambient occlusion method for shadow estimation in the 3DGS pipeline, which can be integrated into optimization without extra components while maintaining efficiency. Our method outperforms existing method, offering an effective solution for 3D human modeling.

However, our deformation prior extraction relies on predefined priors, which not capture diverse information across clothed human body. Future work could incorporate richer physical knowledge or priors from large-language model [1, 38]. Moreover, the radius optimization could theoretically be integrated into StyleUNets for greater flexibility, but we leave this exploration for future work.

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023. 8
- [2] Thiemo Alldieck, Marcus Magnor, Weipeng Xu, Christian Theobalt, and Gerard Pons-Moll. Detailed human avatars from monocular video. In *2018 International Conference on 3D Vision (3DV)*, pages 98–109. IEEE, 2018. 2
- [3] Thiemo Alldieck, Marcus Magnor, Weipeng Xu, Christian Theobalt, and Gerard Pons-Moll. Video based reconstruction of 3d people models. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 8387–8397, 2018. 2
- [4] Louis Bavoil and Miguel Sainz. Screen space ambient occlusion. *NVIDIA developer information: <http://developers.nvidia.com>*, 6(2):6, 2008. 4
- [5] Brent Burley and Walt Disney Animation Studios. Physically-based shading at disney. In *Acm siggraph*, pages 1–7. vol. 2012, 2012. 1, 4
- [6] Andrei Burov, Matthias Nießner, and Justus Thies. Dynamic surface function networks for clothed human bodies. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10754–10764, 2021. 2
- [7] Diogo Carbonera Luvizon, Vladislav Golyanik, Adam Kortylewski, Marc Habermann, and Christian Theobalt. Relightable neural actor with intrinsic decomposition and pose control. In *European Conference on Computer Vision*, pages 465–483. Springer, 2024. 2
- [8] Hongze Chen, Zehong Lin, and Jun Zhang. Gi-gs: Global illumination decomposition on gaussian splatting for inverse rendering. *arXiv preprint arXiv:2410.02619*, 2024. 2
- [9] Yue Chen, Xuan Wang, Xingyu Chen, Qi Zhang, Xiaoyu Li, Yu Guo, Jue Wang, and Fei Wang. Uv volumes for real-time rendering of editable free-view human performance. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16621–16631, 2023. 2
- [10] Yushuo Chen, Zerong Zheng, Zhe Li, Chao Xu, and Yebin Liu. Meshavatar: Learning high-quality triangular human avatars from multi-view videos. In *European Conference on Computer Vision*, pages 250–269. Springer, 2024. 2, 5, 6, 7
- [11] Zhaoxi Chen and Ziwei Liu. Relighting4d: Neural relightable human from videos. In *European conference on computer vision*, pages 606–623. Springer, 2022. 2, 5, 6, 7
- [12] Seonghwa Choi, Moonkyeong Choi, Mingyu Jang, Jaekyung Kim, Jianfei Cai, Wen-Huang Cheng, and Sanghoon Lee. Relightable and dynamic gaussian avatar reconstruction from monocular video. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pages 7405–7414, 2025. 1, 2
- [13] Pinxuan Dai, Jiamin Xu, Wenxiang Xie, Xinguo Liu, Huamin Wang, and Weiwei Xu. High-quality surface reconstruction using gaussian surfels. In *ACM SIGGRAPH 2024 conference papers*, pages 1–11, 2024. 2, 3, 8, 12
- [14] Paul Debevec, Tim Hawkins, Chris Tchou, Haarm-Pieter Duiker, Westley Sarokin, and Mark Sagar. Acquiring the reflectance field of a human face. In *Proceedings of the 27th annual conference on Computer graphics and interactive techniques*, pages 145–156, 2000. 1
- [15] Jian Gao, Chun Gu, Youtian Lin, Zhihao Li, Hao Zhu, Xun Cao, Li Zhang, and Yao Yao. Relightable 3d gaussians: Realistic point cloud relighting with brdf decomposition and ray tracing. In *European Conference on Computer Vision*, pages 73–89. Springer, 2024. 2
- [16] Kaiwen Guo, Peter Lincoln, Philip Davidson, Jay Busch, Xueming Yu, Matt Whalen, Geoff Harvey, Sergio Orts-Escolano, Rohit Pandey, Jason Dourgarian, et al. The relightables: Volumetric performance capture of humans with realistic relighting. *ACM Transactions on Graphics (ToG)*, 38(6):1–19, 2019. 1
- [17] Marc Habermann, Lingjie Liu, Weipeng Xu, Gerard Pons-Moll, Michael Zollhoefer, and Christian Theobalt. Hdhumans: A hybrid approach for high-fidelity digital humans. *Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 6(3):1–23, 2023. 2
- [18] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017. 4
- [19] Liangxiao Hu, Hongwen Zhang, Yuxiang Zhang, Boyao Zhou, Boning Liu, Shengping Zhang, and Liqiang Nie. Gaussianavatar: Towards realistic human avatar modeling from a single video via animatable 3d gaussians. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 634–644, 2024. 2
- [20] Shoukang Hu, Tao Hu, and Ziwei Liu. Gauhuman: Articulated gaussian splatting from monocular human videos. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 20418–20431, 2024. 2
- [21] Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, and Shenghua Gao. 2d gaussian splatting for geometrically accurate radiance fields. In *ACM SIGGRAPH 2024 conference papers*, pages 1–11, 2024. 8
- [22] Umar Iqbal, Akin Caliskan, Koki Nagano, Sameh Khamis, Pavlo Molchanov, and Jan Kautz. Rana: Relightable articulated neural avatars. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 23142–23153, 2023. 2
- [23] Mustafa Işık, Martin Rünz, Markos Georgopoulos, Taras Khakhulin, Jonathan Starck, Lourdes Agapito, and Matthias Nießner. Humanrf: High-fidelity neural radiance fields for humans in motion. *ACM transactions on graphics (TOG)*, 42(4):1–12, 2023. 4, 5, 6
- [24] Chaonan Ji, Tao Yu, Kaiwen Guo, Jingxin Liu, and Yebin Liu. Geometry-aware single-image full-body human relighting. In *European Conference on Computer Vision*, pages 388–405. Springer, 2022. 2
- [25] Yujiao Jiang, Qingmin Liao, Xiaoyu Li, Li Ma, Qi Zhang, Chaopeng Zhang, Zongqing Lu, and Ying Shan. Uv gaussians: Joint learning of mesh deformation and gaussian textures for human avatar modeling. *Knowledge-Based Systems*, 320:113470, 2025. 2

- [26] Zeren Jiang, Shaofei Wang, and Siyu Tang. Dnf-avatar: Distilling neural fields for real-time animatable avatar relighting. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 6383–6394, 2025. 1, 2, 8
- [27] Yoshihiro Kanamori and Yuki Endo. Relighting humans: occlusion-aware inverse rendering for full-body human images. *ACM Transactions on Graphics (TOG)*, 37(6):1–11, 2018. 2
- [28] Timothy L Kay and James T Kajiya. Ray tracing complex scenes. *ACM SIGGRAPH computer graphics*, 20(4):269–278, 1986. 8
- [29] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkuehler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics (TOG)*, 42(4):1–14, 2023. 1, 2
- [30] Muhammed Kocabas, Jen-Hao Rick Chang, James Gabriel, Oncel Tuzel, and Anurag Ranjan. Hugs: Human gaussian splats. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 505–515, 2024. 2
- [31] Youngjoong Kwon, Lingjie Liu, Henry Fuchs, Marc Habermann, and Christian Theobalt. Deliffas: Deformable light fields for fast avatar synthesis. *Advances in neural information processing systems*, 36:40944–40962, 2023. 2
- [32] Mengtian Li, Shengxiang Yao, Chen Kai, Zhifeng Xie, Keyu Chen, and Yu-Gang Jiang. Anigaussian: Animatable gaussian avatar with pose-guided deformation. *arXiv preprint arXiv:2502.19441*, 2025. 2
- [33] Zhe Li, Yipengjing Sun, Zerong Zheng, Lizhen Wang, Shengping Zhang, and Yebin Liu. Animatable and relightable gaussians for high-fidelity human avatar modeling. *arXiv preprint arXiv:2311.16096*, 2023. 1, 2, 6
- [34] Zhe Li, Zerong Zheng, Yuxiao Liu, Boyao Zhou, and Yebin Liu. Posevocab: Learning joint-structured pose embeddings for human avatar modeling. In *ACM SIGGRAPH 2023 conference proceedings*, pages 1–11, 2023. 2, 5
- [35] Zhe Li, Zerong Zheng, Lizhen Wang, and Yebin Liu. Animatable gaussians: Learning pose-dependent gaussian maps for high-fidelity human avatar modeling. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 19711–19722, 2024. 2, 3, 4, 5, 12
- [36] Zhihao Liang, Qi Zhang, Ying Feng, Ying Shan, and Kui Jia. Gs-ir: 3d gaussian splatting for inverse rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 21644–21653, 2024. 2, 8
- [37] Wenbin Lin, Chengwei Zheng, Jun-Hai Yong, and Feng Xu. Relightable and animatable neural avatars from videos. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 3486–3494, 2024. 1, 2
- [38] Aixiu Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, et al. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024. 8
- [39] Lingjie Liu, Marc Habermann, Viktor Rudnev, Kripasindhu Sarkar, Jiatao Gu, and Christian Theobalt. Neural actor: Neural free-view synthesis of human actors with pose control. *ACM transactions on graphics (TOG)*, 40(6):1–16, 2021. 2
- [40] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J Black. Smpl: a skinned multi-person linear model. *ACM Transactions on Graphics (TOG)*, 34(6):1–16, 2015. 1, 2
- [41] Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. *Communications of the ACM*, 65(1):99–106, 2021. 1
- [42] Rohit Pandey, Sergio Orts-Escolano, Chloe Legendre, Christian Haene, Sofien Bouaziz, Christoph Rhemann, Paul E Debevec, and Sean Ryan Fanello. Total relighting: learning to relight portraits for background replacement. *ACM Trans. Graph.*, 40(4):43–1, 2021. 2
- [43] Haokai Pang, Heming Zhu, Adam Kortylewski, Christian Theobalt, and Marc Habermann. Ash: Animatable gaussian splats for efficient and photoreal human rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1165–1175, 2024. 2
- [44] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style, high-performance deep learning library. *Advances in neural information processing systems*, 32, 2019. 4
- [45] Georgios Pavlakos, Vasileios Choutas, Nima Ghorbani, Timo Bolkart, Ahmed A. A. Osman, Dimitrios Tzionas, and Michael J. Black. Expressive body capture: 3D hands, face, and body from a single image. In *Proceedings IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, pages 10975–10985, 2019. 1, 2, 3, 12
- [46] Zhiyin Qian, Shaofei Wang, Marko Mihajlovic, Andreas Geiger, and Siyu Tang. 3dgs-avatar: Animatable avatars via deformable 3d gaussian splatting. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5020–5030, 2024. 2, 5, 6
- [47] Why Physically-Based Rendering. Physically-based rendering. *Procedia IUTAM*, 13(127-137):3, 2015. 1, 3, 12
- [48] Shunsuke Saito, Gabriel Schwartz, Tomas Simon, Junxuan Li, and Giljoo Nam. Relightable gaussian codec avatars. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 130–141, 2024. 2
- [49] Tiancheng Sun, Jonathan T Barron, Yun-Ta Tsai, Zexiang Xu, Xueming Yu, Graham Fyfe, Christoph Rhemann, Jay Busch, Paul Debevec, and Ravi Ramamoorthi. Single image portrait relighting. *ACM Trans. Graph.*, 38(4):1–12, 2019. 2
- [50] Bruce Walter, Stephen R Marschner, Hongsong Li, and Kenneth E Torrance. Microfacet models for refraction through rough surfaces. *Rendering techniques*, 2007:18th, 2007. 2, 12
- [51] Lizhen Wang, Xiaochen Zhao, Jingxiang Sun, Yuxiang Zhang, Hongwen Zhang, Tao Yu, and Yebin Liu. Styleavatar: Real-time photo-realistic portrait avatar from a single video. In *ACM SIGGRAPH 2023 Conference Proceedings*, pages 1–10, 2023. 3
- [52] Shaofei Wang, Katja Schwarz, Andreas Geiger, and Siyu Tang. Arah: Animatable volume rendering of articulated hu-

- man sdf. In *European conference on computer vision*, pages 1–19. Springer, 2022. [2](#)
- [53] Shaofei Wang, Božidar Antić, Andreas Geiger, and Siyu Tang. Intrinsicavatar: Physically based inverse rendering of dynamic humans from monocular videos via explicit ray tracing. In *Proceedings IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2024. [2](#), [8](#)
- [54] Shaofei Wang, Tomas Simon, Igor Santesteban, Timur Bagautdinov, Junxuan Li, Vasu Agrawal, Fabian Prada, Shou-I Yu, Pace Nalbone, Matt Gramlich, et al. Relightable full-body gaussian codec avatars. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*, pages 1–12, 2025. [1](#), [2](#)
- [55] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing*, 13(4):600–612, 2004. [4](#)
- [56] Zhibo Wang, Xin Yu, Ming Lu, Quan Wang, Chen Qian, and Feng Xu. Single image portrait relighting via explicit multiple reflectance channel modeling. *ACM Transactions on Graphics (ToG)*, 39(6):1–13, 2020. [2](#)
- [57] Jiacheng Wu, Ruiqi Zhang, Jie Chen, and Hui Zhang. Fast and physically-based neural explicit surface for relightable human avatars. In *2025 IEEE International Conference on Multimedia and Expo (ICME)*, pages 1–6. IEEE, 2025. [1](#)
- [58] Donglai Xiang, Timur Bagautdinov, Tuur Stuyck, Fabian Prada, Javier Romero, Weipeng Xu, Shunsuke Saito, Jingfan Guo, Breannan Smith, Takaaki Shiratori, et al. Dressing avatars: Deep photorealistic appearance for physically simulated clothing. *ACM Transactions on Graphics (TOG)*, 41(6):1–15, 2022. [2](#)
- [59] Junjin Xiao, Qing Zhang, Zhan Xu, and Wei-Shi Zheng. Neca: Neural customizable human avatar. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 20091–20101, 2024. [1](#), [2](#)
- [60] Feng Xu, Yebin Liu, Carsten Stoll, James Tompkin, Gaurav Bharaj, Qionghai Dai, Hans-Peter Seidel, Jan Kautz, and Christian Theobalt. Video-based characters: creating new human performances from a multi-view video database. *TOG*, 30(4):1–10, 2011. [2](#)
- [61] Zhen Xu, Sida Peng, Chen Geng, Linzhan Mou, Zihan Yan, Jiaming Sun, Hujun Bao, and Xiaowei Zhou. Relightable and animatable neural avatar from sparse-view video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 990–1000, 2024. [1](#), [2](#)
- [62] Lingzhe Zeng, Wensheng Li, Rongbin Zheng, and Chengying Gao. Rega: Relighting dynamic gaussian avatars from sparse views. *IEEE Transactions on Circuits and Systems for Video Technology*, 2025. [2](#)
- [63] Youyi Zhan, Tianjia Shao, He Wang, Yin Yang, and Kun Zhou. Interactive rendering of relightable and animatable gaussian avatars. *IEEE Transactions on Visualization and Computer Graphics*, 2025. [1](#), [2](#), [8](#)
- [64] Youyi Zhan, He Wang, Tianjia Shao, and Kun Zhou. High-fidelity mobile avatars with pruned local blendshapes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 32345–32356, 2026. [5](#), [6](#)
- [65] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 586–595, 2018. [4](#)
- [66] Yiqun Zhao, Chenming Wu, Binbin Huang, Yihao Zhi, Chen Zhao, Jingdong Wang, and Shenghua Gao. Surfel-based gaussian inverse rendering for fast and relightable dynamic human reconstruction from monocular videos. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025. [1](#)
- [67] Shunyuan Zheng, Boyao Zhou, Ruizhi Shao, Boning Liu, Shengping Zhang, Liqiang Nie, and Yebin Liu. Gps-gaussian: Generalizable pixel-wise 3d gaussian splatting for real-time human novel view synthesis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 19680–19690, 2024. [2](#)
- [68] Zerong Zheng, Han Huang, Tao Yu, Hongwen Zhang, Yandong Guo, and Yebin Liu. Structured local radiance fields for human avatar modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15893–15903, 2022. [2](#)
- [69] Zerong Zheng, Xiaochen Zhao, Hongwen Zhang, Boning Liu, and Yebin Liu. Avatarrex: Real-time expressive full-body avatars. *ACM Transactions on Graphics (TOG)*, 42(4):1–19, 2023. [2](#), [4](#), [5](#), [6](#)
- [70] Hao Zhou, Sunil Hadap, Kalyan Sunkavalli, and David W Jacobs. Deep single-image portrait relighting. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 7194–7202, 2019. [2](#)
- [71] Wojciech Zielonka, Timur Bagautdinov, Shunsuke Saito, Michael Zollhöfer, Justus Thies, and Javier Romero. Drivable 3d gaussian avatars. In *2025 International Conference on 3D Vision (3DV)*, pages 979–990. IEEE, 2025. [2](#)
- [72] Matthias Zwicker, Hanspeter Pfister, Jeroen Van Baar, and Markus Gross. Ewa volume splatting. In *Proceedings Visualization, 2001. VIS'01.*, pages 29–538. IEEE, 2001. [3](#), [4](#)

Supplemental Document

6. Implementation Details

All models are implemented in PyTorch and trained on a single NVIDIA RTX 3090 GPU with 24 GB of memory. The rasteriser, the cubemap filtering and the differentiable ambient occlusion formulation are custom CUDA extensions. The split-sum pre-filtering uses the same cosine and GGX kernels as the public `nvdiffrrec` release. The PCA operation for out-of-distribution pose [35] is also performed on position map and deformation priors map for more faithful reconstruction.

7. Detailed Deformation Feature Maps

As mentioned in Sec. 3.2, we define the deformation feature maps by utilizing two distance: a) the distance before and after deformation $\mathbf{d}$, and b) the distance to the nearest SMPL-X [45] vertex $\mathbf{s}$. For the two distances, we encode their magnitude and orientation into the feature map $\mathbf{M}_{def}^f$ and $\mathbf{M}_{def}^b$ respectively. However, for distance $\mathbf{s}$, it measures the error of the weights in the LBS deformation process. Therefore, we believe that its magnitude is more valuable. Therefore, the final encoded information includes the magnitude and orientation of $\mathbf{d}$ and the magnitude of $\mathbf{s}$,

$$\{\mathbf{d}, \mathbf{s}\} = \text{Concate} \left(\|\mathbf{d}\|, \frac{\mathbf{d}}{\|\mathbf{d}\|}, \|\mathbf{s}\| \right), \quad (21)$$

where Concate means concatenate operation.

8. Detailed Splatting Formulation

We employ $\text{Splat}(\cdot; \cdot)$ introduced in GaussianSrfel [13], which render color $\hat{C}$, normal $\hat{N}$, depth $\hat{D}$ maps from a set of surfel. We extend the formulation in manuscript to render albedo $\hat{A}$, roughness $\hat{H}$, position $\hat{P}$, radius $\hat{R}$, perturbed radius $\hat{R}_+$, $\hat{R}_-$, sample importance I , and SSAO bone-aware direction Q maps. Therefore, the final splatting formulation is defined as:

$$\begin{aligned} &\hat{C}, \hat{D}, \hat{N}, \hat{A}, \hat{H}, \hat{P}, \hat{R}, \\ &\hat{R}_+, \hat{R}_-, I, Q = \text{Splat}(\mathbf{c}, \mathbf{a}, \rho; \mu_o, \Sigma_o, \mathbf{o}), \end{aligned} \quad (22)$$

we will discuss the sample importance map I and SSAO direction map Q in Sec. 10. The rendered albedo and roughness map is splatted without the influence of background, which avoids training failure caused by background noise during deferred shading.

9. Detailed Physically-Based Shading

Specifically, we decompose the BRDF term into diffuse and specular components:

$$f_r(\omega_i, \omega_o) = f_{\text{diff}}(\omega_i) + f_{\text{spec}}(\omega_i, \omega_o), \quad (23)$$

The diffuse color is computed by:

$$C_{\text{diff}} = \frac{\mathbf{a}}{\pi} \cdot \int_{\Omega} L_i(\omega_i)(\omega_i \cdot \mathbf{n}) d\omega_i, \quad (24)$$

For specular component, we utilize SplitSum approximation to separate the light and BRDF term in the integrals. Therefore, the specular color is computed by:

$$\begin{aligned} C_{\text{spec}} &\approx \int_{\Omega} L_i(\omega_i) D(\omega_i, \omega_o)(\omega_i, n) d\omega_i \cdot \\ &\int_{\Omega} f_{\text{spec}}(\omega_i, \omega_o) \cdot (\omega_i \cdot n) d\omega \end{aligned} \quad (25)$$

where $D(\omega_i, \omega_o)$ denotes normal distribution function. We discuss the terms D and f_{spec} in the appendix. The diffuse and specular components are finally combined to obtain the shaded color before accounting for visibility:

$$\hat{C}_{\text{shade}} = C_{\text{diff}} + C_{\text{spec}}, \quad (26)$$

As mentioned in Sec. 3.3, we employ the rendering equation [47] on rendered maps, and the BRDF term $f_r(\omega_i, \omega_o)$ is decomposed into diffuse and specular components. The diffuse component is defined as:

$$f_{\text{diff}}(\omega_i) = (1 - m) \cdot \frac{\mathbf{a}}{\pi}, \quad (27)$$

We simplify the Eq. 27 by set metallic equals 0 to get the final diffuse BRDF term $f_{\text{diff}} = \frac{\mathbf{a}}{\pi}$.

For specular component f_{spec} , we define it using normal distribution function D , Fresnel term F , and geometry term G , which are calculated in [50],

$$f_{\text{spec}}(\omega_i, \omega_o) = \frac{D \cdot F \cdot G}{4 \cdot (\omega_i \cdot n) \cdot (\omega_o \cdot n)}, \quad (28)$$

where n denotes the normal of current pixel.

10. Detailed Differentiable Ambient Occlusion

This section provides the detailed formulation of the differentiable SSAO module introduced in Sec. 3.4. The central objective is to enable gradient-based optimization of the occlusion radius while preserving the efficient CUDA implementation of the original SSAO operator.

10.1. Body-Part Specific Radius Parameterization

We divide the human body into nine semantic parts and generate a body set:

$$\mathcal{B} = \{\text{torso, hip, knee, foot, head, shoulder, elbow, wrist, hand}\}, \quad (29)$$

each body part $b \in \mathcal{B}$ is assigned an initial radius r_g^{init} and a learnable offset δ_g . We have discuss how to obtain radii of each surfel in Eq. (13 ~ 15). Finally, we assign the radius attribute r_i to each surfel and make it participate in splatting to obtain the rendered radius map $\hat{R}$.

10.2. Custom Backward Propagation

Given an upstream gradient $\frac{\partial \mathcal{L}}{\partial \hat{O}}$, the custom backward operation computes as,

$$\frac{\partial \mathcal{L}}{\partial \hat{R}} = \frac{\partial \mathcal{L}}{\partial \hat{O}} \cdot \frac{\partial \hat{O}}{\partial \hat{R}}, \quad (30)$$

the resulting gradient is propagated through the differentiable baseline radius branch:

$$\frac{\partial \mathcal{L}}{\partial \hat{R}} \rightarrow \frac{\partial \mathcal{L}}{\partial \{r_i\}} \rightarrow \frac{\partial \mathcal{L}}{\partial \{r_g\}} \rightarrow \frac{\partial \mathcal{L}}{\partial \{\delta_g\}_{g=1}^9}, \quad (31)$$

two perturbed positive and negative radius branches are detached from the computational graph and therefore do not receive gradients. Likewise, the normal, depth, bone-aware direction, and importance maps are treated as fixed inputs to the differentiable SSAO module:

$$\frac{\partial \hat{O}}{\partial \hat{N}} = \frac{\partial \hat{O}}{\partial \hat{D}} = \frac{\partial \hat{O}}{\partial Q} = \frac{\partial \hat{O}}{\partial I} = \mathbf{0}, \quad (32)$$

consequently, the proposed module provides a gradient path only for the body-part specific radius parameters.

10.3. Bone-Aware Direction Map

To guide the SSAO sampling direction, we introduce a bone-aware direction map Q that incorporates pose-dependent skeletal information into the sampling process. For each surfel primitive i , we first compute its bone-aware direction by blending the joint frame axes with the corresponding LBS weights:

$$\mathbf{q}_i^{\text{bone}} = \text{normalize} \left(\sum_{j=1}^J w_{ij} \mathbf{V}_j \mathbf{e}_y \right), \quad (33)$$

where w_{ij} denotes the LBS weight between surfel i and joint j , $\mathbf{e}_y = (0, 1, 0)$ is the canonical up axis, and $\mathbf{V}_j = \mathbf{R}_j \mathbf{R}_j^{\text{rest}^{-1}}$ is the pose-dependent rotation of joint j , expressed as the joint's canonical-to-live rotation $\mathbf{R}_j$ composed with the inverse of its rotation $\mathbf{R}_j^{\text{rest}}$ in the canonical pose.

The direction $\mathbf{q}_i^{\text{bone}}$ is produced in world space and is rotated into camera space before being splatted, so that it is expressed in the same frame as the camera-space surfel normal $\mathbf{n}_i$. Since SSAO sampling is performed around the surface tangent plane, we project the bone direction onto the tangent space of each surfel:

$$\mathbf{t}'_i = \mathbf{q}_i^{\text{bone}} - (\mathbf{q}_i^{\text{bone}} \cdot \mathbf{n}_i) \mathbf{n}_i, \quad (34)$$

$$\mathbf{t}_i = \text{normalize}(\mathbf{t}'_i), \quad \mathbf{b}_i = \mathbf{n}_i \times \mathbf{t}_i, \quad (35)$$

the tangent basis is then defined as:

$$\mathbf{Q}_i = [\mathbf{t}_i, \mathbf{b}_i, \mathbf{n}_i], \quad (36)$$

Similar to other surfel attributes, the direction field $\{\mathbf{q}_i^{\text{bone}}\}$ is splatted into image space using $\text{Splat}(\cdot; \cdot)$ to obtain the bone-aware direction map Q .

10.4. Sample Importance Map

Our screen-space ambient occlusion solver marches a fixed set of hemisphere directions on a uniform angular grid, so that every pixel receives the same number of occlusion rays. This uniform allocation is a poor fit for an articulated human body, whose parts span more than an order of magnitude in size: because a ray-marching step maps to a screen-space displacement proportional to the surface depth, a density tuned to resolve occlusion on the trunk wastes samples on its wide variation, while the same density is far too sparse to resolve the narrow occlusion features of fingers and wrists. We therefore introduce a sample importance map I that allocates SSAO sampling density according to the local body-part scale, and is cheap to evaluate because it reuses the LBS weights and joints already available from skinning.

For each surfel i we first compute its skeletal reference position by LBS-weighted interpolation of joint locations:

$$\mathbf{x}_i^{\text{bone}} = \sum_{j=1}^J w_{ij} \mathbf{x}_j, \quad (37)$$

where $\mathbf{p}_j$ denotes the position of the j -th joint in the posed frame, w_{ij} is the LBS weight of surfel i at joint j , and J is the number of joints.

A uniform distance decay would treat all joints alike, but the relevant length scale differs by an order of magnitude between, e.g., a finger joint and the pelvis. We therefore associate each joint j with a nominal body-part radius r_j and define the importance of a surfel as the ratio between its blended body-part scale and the per-joint scale of the reference position:

$$I_i = \frac{\sum_{j=1}^J w_{ij} e^{-\|\mathbf{x}_i - \mathbf{p}_j\|_{2/\sigma} r_j}}{\sum_{j=1}^J w_{ij} r_j} \in [0, 1], \quad (38)$$

where $\mathbf{x}_i$ is the position of surfel i , and $\sigma = 0.15 \text{ m}$ is a distance fall-off radius. The numerator is an exponentially distance-weighted average of the per-joint radii over the joints that actually influence the surfel, so a surfel lying on a thin structure such as a finger inherits the small radius of its governing joint and obtains $I_i \rightarrow 1$, whereas a surfel on a thick structure such as the torso is dominated by the large radii of the trunk joints and obtains a small I_i . The exponential stencil makes the transition between neighbouring body parts smooth, so no part segmentation is required; because it appears in both numerator and denominator, the ratio is also independent of the absolute scale of the joint radii.

Similar to other surfel attributes, the importance values are packed into the SSAO feature channel of the surfel and

splatted using $\text{Splat}(\cdot; \cdot)$. The obtained importance map I is used to adjust the angular sampling density of SSAO:

$$\delta_{\text{adp}} = \frac{\delta}{1 + \alpha I}, \quad (39)$$

where $\delta = 0.0625$ is the default angular sample interval, and $\alpha = 3$ is a weight that controls the influence of the importance map on the sampling density. Allocating a full unit of angular budget to the most articulated regions yields a denser step $\delta_{\text{adp}} \in [\delta/4, \delta]$, i.e. up to four times more sampled directions on thin appendages, while leaving the sampling on the torso unchanged. The kernel clamps the incoming importance to $[0, 1]$ before applying (39) to keep the number of sampled directions bounded in all cases.

Table 4. **Loss terms and weights.** The depth normal consistency weight ramps up from 0.025 to 0.125 during training.

Loss	Symbol	Weight	Stage
Photometric L1	$\mathcal{L}_1$	1.0	Rad., PBR
Perceptual (LPIPS)	$\mathcal{L}_{\text{lpiPs}}$	0.1	Rad., PBR
Offset regularisation	$\mathcal{L}_{\text{reg}}$	0.005	Rad., PBR
Depth normal consistency	$\mathcal{L}_{\text{cons}}$	0.025	Rad., PBR
Curvature smoothness	$\mathcal{L}_{\text{curv}}$	0.005	Rad., PBR
Normal deformation	$\mathcal{L}_{\text{nd}}$	0.004	Rad.
BRDF TV	$\mathcal{L}_{\text{TV}}$	0.005	PBR
Environment prior	$\mathcal{L}_{\text{light}}$	0.05	PBR

10.5. Detailed Training Losses

The training process include two stage, In the first stage, we use the directional color and reconstruct the geometry of the human avatar. The training loss consist of radiance loss, geometry loss, normal deformation loss, and regularization. The radiance loss includes L1 and lpips loss,

$$\mathcal{L}_{\text{rad}} = \mathcal{L}_1(C, \hat{C}) + \mathcal{L}_{\text{lpiPs}}(C, \hat{C}), \quad (40)$$

and geometry loss is defined as:

$$\mathcal{L}_{\text{geo}} = \mathcal{L}_{\text{cons}}(\hat{D}, \hat{N}) + \mathcal{L}_{\text{curv}}, \quad (41)$$

We have discuss the definition of normal deformation loss $\mathcal{L}_{\text{nd}}$ and regularize the offset,

$$\mathcal{L}_{\text{reg}} = \|\Delta\mu\|_2^2, \quad (42)$$

in the second stage, we estimate BRDF materials and light and add extended training loss, including shading loss, TV smooth loss, and regularization of light.

$$\mathcal{L}_{\text{shaded}} = \mathcal{L}_1(C, \hat{C}_{\text{pbr}}) + \mathcal{L}_{\text{lpiPs}}(C, \hat{C}_{\text{pbr}}), \quad (43)$$

We employ the TV smooth loss $\mathcal{L}_{\text{TV}}$ on rendered albedo and rendered roughness map and environment light, and enforce the light tend to white light,

$$\mathcal{L}_{\text{light}} = \frac{1}{3} \sum_{i=1}^3 \left\| l_i - \frac{1}{3} \sum_{j=1}^3 l_j \right\|. \quad (44)$$

we use $\mathcal{L}_{\text{rad}}$ in the first stage and switched to $\mathcal{L}_{\text{shaded}}$ in the second stage, while also adding $\mathcal{L}_{\text{TV}}$ and $\mathcal{L}_{\text{light}}$.